

Latency-aware PM2.5 interval forecasting under simulated measurement and feedback delays: a reproducible Beijing benchmark

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Abstract

Observation latency restricts both forecast inputs and responses available for recalibration. This retrospective Beijing benchmark evaluates 24-hour PM2.5 forecasts using 420,768 station-hour records and 102,465 observed final-period targets. Two distinct objectives organize the evidence. The locally prespecified analysis compares rolling calibration with valid feedback against static unmatched calibration under a fixed predictor. At a simulated 72-hour delay and 90% nominal level, rolling reduces mean interval score from 377.14 to 328.01, with coverage 88.32%; its paired difference interval excludes zero with seven-day blocks but includes zero with 14- and 28-day blocks. A post-review exploration evaluates alternative pipelines, including CQR with delay-specific training and an ACI-inspired daily delayed-feedback heuristic (ADH). At that same delay and nominal level, CQR has the lowest observed score among evaluated pipelines (258.40), coverage 88.91% and mean width 182.72, whereas ADH is closer to nominal coverage (89.78%) with greater width (203.17) and score (317.48). CQR coverage has a pointwise 14-day block-bootstrap interval of [86.51, 91.18]%; inclusion of 90% does not establish calibration. The CQR–rolling score reduction of 21.2% is exploratory and compares complete pipelines, not isolated calibration effects. Synthetic heterogeneous delays, missingness diagnostics and an earlier Beijing backtest assess sensitivity rather than independent confirmation. The contribution is a reproducible information-set benchmark, not a new algorithm, universal coverage guarantee or demonstrated transfer to other cities.

Keywords: PM2.5 forecasting, observation latency, uncertainty quantification, temporal validation, conformal calibration, environmental informatics

1 Introduction

Air-quality forecasts are made with the measurements available at their issue time. Retrospective datasets, in contrast, often provide a completed archive without an explicit record of publication latency. A forecast evaluation can therefore inadvertently grant access to measurements, or to forecast errors, before they would have become available. This distinction matters when prediction intervals are recalibrated online: the observed response needed to score an earlier forecast may itself be delayed.

Conformal inference offers a general route from predictive models to uncertainty sets. Adaptive conformal inference addresses distribution shift (Gibbs and Candès 2021), and subsequent work studies its behaviour and aggregation for dependent series (Zaffran et al. 2022). Sequential predictive conformal inference explicitly exploits dependence in conformity scores (Xu and Xie 2023). Missing covariates introduce another difficulty: average coverage can conceal poor coverage for particular missingness patterns (Zaffran et al. 2023). These established contributions motivate an empirical evaluation that keeps information availability explicit.

The present study uses the Beijing multi-site air-quality archive (Chen 2017; Zhang et al. 2017) to ask how point and interval forecasts change when the PM2.5 channel arrives late. The analysis does not propose a new conformal algorithm. Its contribution is a reproducible, temporally separated benchmark in which the same simulated delay governs both feature construction and eligibility of residuals for recalibration. An intentionally infeasible feedback rule, which ignores the latter restriction, is reported separately as a diagnostic.

Three questions guide the experiment. How much do simple point predictors deteriorate under increasing delay? Does matching the calibration information state or updating calibration over time improve interval performance? Are the resulting conclusions stable across locations, calendar periods and dependence-sensitive uncertainty calculations? Interval score is used alongside coverage and width to avoid rewarding intervals solely for being wide (Gneiting and Raftery 2007).

1.1 Related work and methodological positioning

Environmental forecasting already includes comparisons of conditional quantile models for NO₂ (Vasseur and Aznarte 2021) and multi-step prediction intervals for locally stationary pollutant series (Li et al. 2022). These studies are closer methodological precedents than generic artificial-intelligence reviews. The present question concerns a separate operational restriction: which inputs and forecast errors have arrived at issue time? Neither reference establishes the behaviour of the delayed-information procedures evaluated here.

The foundations of conformal prediction separate the construction of uncertainty sets from the choice of point learner (Vovk et al. 2005; Angelopoulos and Bates 2023). Regression methods include split-conformal inference and resampling-based alternatives such as the jackknife+ (Lei et al. 2018; Barber et al. 2021). Work beyond

exchangeability explicitly examines departures from the assumptions supporting classical guarantees (Barber et al. 2023); a recent survey organizes further developments from a data perspective (Zhou et al. 2026). These references motivate the benchmark but do not supply an automatic coverage theorem for its pooled, nonstationary environmental series. The original rolling residual rule is distinct from ACI, SPCI and jackknife+. The extension implements CQR and an ACI-inspired daily delayed-feedback heuristic (ADH), without attributing the original ACI theorem to that modification.

Evaluation design is a separate source of uncertainty. Forecasting research emphasizes out-of-sample assessment aligned with deployment (Tashman 2000). Cross-validation can be valid for particular autoregressive settings under appropriate conditions (Bergmeir et al. 2018); this does not justify random splitting for every dependent dataset. Structured environmental data require a splitting strategy that reflects the intended prediction task (Roberts et al. 2017). Selection on the same validation results used to estimate performance can induce optimism (Varma and Simon 2006), and information leakage undermines reproducibility in machine-learning-based science (Kapoor and Narayanan 2023). Accordingly, this study separates development, calibration and test periods and makes feedback arrival explicit.

2 Data and chronological design

2.1 Source and audit

The UCI dataset contains hourly air-pollutant and meteorological records at 12 Beijing sites from 1 March 2013 through 28 February 2017. It is distributed under CC BY 4.0 (Chen 2017). The analysis uses the twelve original station files inside the archived PRSA2017 data package. Auxiliary files named `data.csv` and `test.csv` in the outer download are not included. A source checksum and file manifest are retained in the reproducibility materials.

There are 420,768 station-hour records and 18 original columns. The audit found no duplicate station-time keys and no discontinuities in the hourly row grid. PM2.5 is missing in 8,739 records (2.08%). These absences form 2,837 contiguous station-specific episodes; the longest lasts 343 hours at Aotizhongxin, from 17 December 2014 at 10:00 through 31 December 2014 at 16:00. Across the archive, 81 hourly timestamps have no PM2.5 measurement at any of the twelve sites. These are descriptions of recorded availability, not confirmed sensor failures.

Missing-data mechanisms affect the interpretation of analyses on observed records (Rubin 1976). No missing-at-random mechanism is established for this archive. Raw concentrations are preserved without outlier deletion or response imputation. No negative values occur in the six pollutant channels, but this check does not establish instrumental accuracy. Predictor absences remain explicit. Time stamps retain the source clock; no time-zone conversion or real publication-time reconstruction is claimed. Audits of missingness used the complete archive before the experiment, whereas model selection and concentration summaries were restricted to development data.

2.2 Forecast target and sample flow

For site s and issue time t , the response is the hourly concentration $Y_{s,t+24}$, not a daily average. Model development covers March 2013–February 2015, interval calibration covers March 2015–February 2016, and the final test covers March 2016–February 2017. The last 24 issue times per site in each period are excluded so that their response does not cross the period boundary. The first 168 hours of the archive are reserved for constructing the longest historical window. A missing response prevents scoring that origin; a missing input does not automatically remove it.

Table 1 Sample flow. Exclusions are mutually exclusive and applied from left to right. Counts refer to station-specific forecast origins.

Period	Initial	Boundary	History	Response NA	Eligible
Development	210,240	288	2,016	4,226	203,710
Calibration	105,408	288	0	2,120	103,000
Test	105,120	288	0	2,367	102,465

The test includes origins from 1 March 2016 through 27 February 2017; their responses extend through 28 February. Observations from an earlier period can supply historical inputs for a later period, as they would in chronological deployment. No random train-test split is used. The same 102,465 test origins are evaluated under every delay, enabling paired comparisons. This conditions evaluation on originally observed responses and does not establish performance when the unobserved target values are systematically different.

3 Methods

3.1 Information sets and simulated delay

Let $d \in \{0, 6, 24, 72, 168\}$ denote a constant PM2.5 delivery delay in hours. At issue time t , the PM2.5 history is restricted to times $u \leq t - d$. Other pollutant and meteorological channels are assumed to be available at t , retaining their natural missingness. Thus the experiment represents a delayed pollutant channel, not loss of an entire station or a measured latency distribution. The zero-delay condition still contains all naturally missing values.

The baseline feature set contains station identity, current pollutants and meteorology, missingness indicators, PM2.5 lags of 1, 3, 6, 12, 24, 48 and 168 hours, retrospective means, standard deviations and observed counts over 24 and 168 hours, age of the most recent available PM2.5 value, and sinusoidal calendar encodings. Station and wind-direction categories are one-hot encoded using development categories. For $d > 0$, current PM2.5 is hidden, lags shorter than d are hidden, and all dependent

summaries are rebuilt. Specifically, a w -hour summary uses only

$$\{Y_{s,u} : t - w + 1 \leq u \leq t - d, Y_{s,u} \text{ observed}\}. \quad (1)$$

If this set is empty, its mean and standard deviation are missing and its count is zero. Age is computed from the latest observed time no later than $t - d$. Univariate summaries omit missing values and carry their effective count. They never interpolate from future measurements.

3.2 Point predictors and development selection

Ridge regularization and gradient boosting provide established statistical baselines (Hoerl and Kennard 1970; Friedman 2001). The cited boosting framework is conceptual background, not a claim that the original implementation is identical to the histogram estimator used here. In the original experiment, four model families are considered: last-observation persistence; a training mean by station and response hour; Ridge regression; and histogram gradient boosting with absolute-error loss. Persistence falls back to a station-specific median of training responses if no historical value is available. Ridge uses training-median imputation and standardization. Boosting handles numerical missing values directly. All point predictions are truncated below at zero.

Three expanding-origin validation windows cover March–June 2014, July–October 2014 and November 2014–February 2015. Each training set contains only responses dated before the corresponding window. The validation windows also exclude targets beyond their right boundary. Ridge penalties are 1, 10 and 100. Boosting configurations comprise 15 leaves and 100 iterations, 31 leaves and 100 iterations, and 31 leaves and 200 iterations, with learning rate 0.05, minimum leaf size 40 and L_2 regularization 1. Random early stopping is disabled. Configuration selection minimizes the unweighted mean of the three validation MAEs, separately within each family.

The selected models are refitted on the full development period, with no retraining during calibration or test. A fixed boosting model is used for the original interval comparisons, as specified before evaluation. The later exploratory extension is described separately. All four families remain in the point-forecast comparison even when rankings change under delay. The protocol is a locally recorded analysis plan, not a public preregistration.

3.3 Residual calibration with arrival restrictions

For a fixed predictor $\hat{f}_d$ evaluated with delayed features, let

$$R_{s,t,d} = |Y_{s,t+24} - \hat{f}_d(X_{s,t}^{(d)})|. \quad (2)$$

The notation does not imply separate predictor training by delay: the same fitted model receives the differently observed features. Given n available residuals and nominal level $1 - \alpha$, define q_α as the residual of rank $\lceil (n + 1)(1 - \alpha) \rceil$. The prediction

interval is

$$C_{s,t,d} = \left[\max\{0, \hat{Y}_{s,t+24,d} - q_\alpha\}, \hat{Y}_{s,t+24,d} + q_\alpha \right]. \quad (3)$$

Nominal levels are 90% and 95%. Residuals are pooled across stations. This finite-sample quantile convention follows conformal calibration, but dependence and distribution changes preclude asserting an automatic distribution-free guarantee here.

Static unmatched calibration uses residuals computed without an imposed delay, applying the same quantile to every delayed scenario. Static matched calibration recomputes the residuals under the scenario’s feature delay. Both use calibration responses before 23 February 2016 at 00:00, a common cutoff seven days before the first test origin that makes them available under all delays.

Rolling valid calibration is updated once per day. At the start of day a , it uses residuals with response measurement times v satisfying

$$a - d - 30 \text{ days} < v \leq a - d. \quad (4)$$

The associated forecast origins are at most $a - d - 24$ hours. Eligible residuals can come from the calibration period or earlier test forecasts once their responses arrive. Quantiles remain fixed within a day, while point forecasts change hourly. As an implementation diagnostic, a fourth method sets the feedback delay to zero while retaining the delayed input features. For $d > 0$ this rule is operationally infeasible and is excluded from claims about deployable methods.

3.4 Evaluation and uncertainty

Point metrics are MAE and RMSE. Their different loss scales can produce different rankings, making the evaluation criterion part of the forecasting question (Hyndman and Koehler 2006). Interval evaluation reports empirical coverage, mean width, and the interval score

$$\begin{aligned} \text{IS}_\alpha(l, u; y) &= (u - l) + \frac{2}{\alpha}(l - y)\mathbf{1}(y < l) \\ &\quad + \frac{2}{\alpha}(y - u)\mathbf{1}(y > u), \end{aligned} \quad (5)$$

which combines interval width and penalties for observations outside its limits (Gneiting and Raftery 2007). Smaller scores are better. Scores and widths have the same concentration units as the response. Marginal coverage and subgroup coverage are both reported; neither is inferred from interval score alone. This distinction follows the separation between calibration and sharpness in probabilistic forecast assessment (Gneiting et al. 2007). Threshold- and quantile-weighted scores offer additional ways to emphasize particular predictive regions (Gneiting and Ranjan 2011); those alternatives are not computed in this benchmark.

The prespecified primary comparison is rolling valid minus static unmatched interval score at 90% nominal coverage and $d = 72$ hours. Daily sums of paired score differences and daily observation counts are resampled together, retaining all stations within each sampled day. Resampling contiguous blocks is motivated

by the dependent-data bootstrap literature (Kunsch 1989; Lahiri 2003). The stationary bootstrap instead uses random block lengths (Politis and Romano 1994); that alternative is not the procedure implemented here. A circular block bootstrap uses 1,000 replications, seven-day blocks, and percentile limits of 2.5% and 97.5%. Fourteen- and 28-day blocks assess sensitivity. The original generator is NumPy 2.3.5 `default_rng(20261003)`, using PCG64. A single generator advances continuously through block lengths 7, 14 and 28, in that order. Each replicate draws $\lceil N/L \rceil$ start days uniformly with replacement from $0, \dots, N - 1$, expands blocks modulo N , and truncates to N days. Percentiles use `numpy.quantile` with its `linear` interpolation method on unrounded replicates; only displayed limits are rounded to two decimal places. These intervals condition on the fitted models and this network; they omit model-refitting uncertainty. Nonstationarity and the artificial year-end join of a circular bootstrap remain limitations. No hourly independence-based significance test is used.

Circular blocks were chosen in the local protocol to avoid preferentially excluding boundary days from block starts while preserving short sequences and all within-day station dependence. This is a convenient dependent-data approximation, not a model of seasonal stationarity. Its artificial February–March join and mixing of seasonal regimes can distort persistence and percentile limits in either direction. A non-circular moving-block bootstrap removes that join but changes boundary inclusion frequencies; a stationary bootstrap randomizes block lengths and does not itself remove seasonal nonstationarity. Resampling whole years is not informative with one final test year; four unequal, seasonally distinct quarter groups are too few for reliable percentile inference. Leaving a period out addresses stability under a changed evaluation sample rather than the same full-period sampling distribution. These alternatives were not implemented, and block-length sensitivity cannot validate the circular assumption.

3.5 Reproducibility and verification

The accompanying materials include source and protocol hashes, exact partitions, feature specifications, model selection results, fitted estimators, predictions for every delay and tabulated outcomes. Feature checks compare sampled lag values and retrospective means against direct time-key lookups. A separate R implementation recomputes coverage, width and interval score from saved predictions for all 40 method-level combinations, agreeing with Python within 10^{-8} . The primary paired mean is independently reconciled. Python 3.12.14 and R 4.5.2 were used; package versions are recorded with the executable scripts. This verifies arithmetic consistency, not scientific validity of all assumptions.

Local protocol chronology and its limits.

Local session records place the protocol write request at 3 October 2026, 01:12:34.531 UTC; the file creation/write timestamp is 01:12:39.1505342 UTC. The selection file, carrying the protocol hash, was written at 01:15:19.6009483 UTC, before the first recorded final-test command at 01:15:49.776 UTC. The original interval table and final-run manifest were written at 01:17:08.2118407 and 01:17:08.5461841 UTC. The protocol bytes match the initial write payload. Its SHA-256 is:

78c518b2081dc77d4e987595b3fff99aea27958f8e40bb326da86b0e61131af4

Hash comparisons also show that the original interval table, primary bootstrap and daily paired differences match the public v0.2.0 and v0.3.0 snapshots. A supplementary provenance report records these checks. Local logs and filesystem dates are mutable: they document the observed workflow, not an independent timestamp certification or proof that no unrecorded analysis occurred. No separate trusted freeze timestamp exists; “locally prespecified” refers to this documented write-and-hash sequence.

4 Original locally prespecified results

P: Locally prespecified evidence

The original local protocol preceded test evaluation and specified one primary contrast. This is not public preregistration or independent external confirmation. Additional configurations are descriptive; sensitivity analyses are labelled S.

Table 2 Main analysis map: objectives, data and evidential status. P is local prespecification, not independent confirmation.

ID	Status	Data	Main comparison	Outcome
P	Locally prespecified	Final Beijing test	Rolling vs static unmatched, fixed predictor	Score reduction; sensitive to block length
E	Post-review exploratory	Same final test	CQR with delay-specific ADH	Different score/coverage/width trade-offs
S	Sensitivity	Same archive; earlier Beijing period	Block size, pooling, missingness and delay scenarios	Robustness and scope; no external replication

4.1 Temporal selection and point forecasts

The smallest boosting configuration achieves the lowest mean validation MAE (53.00), compared with 60.04 for persistence. Increasing tree complexity does not improve the aggregate selection criterion (Table 3). Ridge with penalty 100 has the lowest validation RMSE, 70.79, although its MAE is higher than boosting. The selection therefore already displays a trade-off between error criteria.

On the final test without additional delay, boosting obtains MAE 51.27, versus 58.12 for persistence and 53.06 for Ridge. Ridge again has smaller RMSE, 73.91 versus 79.53 for boosting. With a 72-hour delay, boosting MAE increases to 55.66 and Ridge MAE is 53.44. Persistence deteriorates to 78.52. These comparisons concern a forecast 24 hours ahead even when its PM2.5 history is several days old. The other pollutant channels remain current by design, and may retain information correlated with the missing channel.

Table 3 Mean development-validation errors across the three temporal windows, in $\mu\text{g}/\text{m}^3$. Configuration selection uses MAE within each family.

Model	MAE	RMSE
Boosting: 15 leaves, 100 iterations	53.00	73.67
Boosting: 31 leaves, 100 iterations	53.28	73.85
Boosting: 31 leaves, 200 iterations	53.72	74.26
Ridge: alpha 100	54.35	70.79
Ridge: alpha 10	54.36	70.80
Ridge: alpha 1	54.36	70.80
Station-hour mean	59.18	75.59
Persistence	60.04	84.45

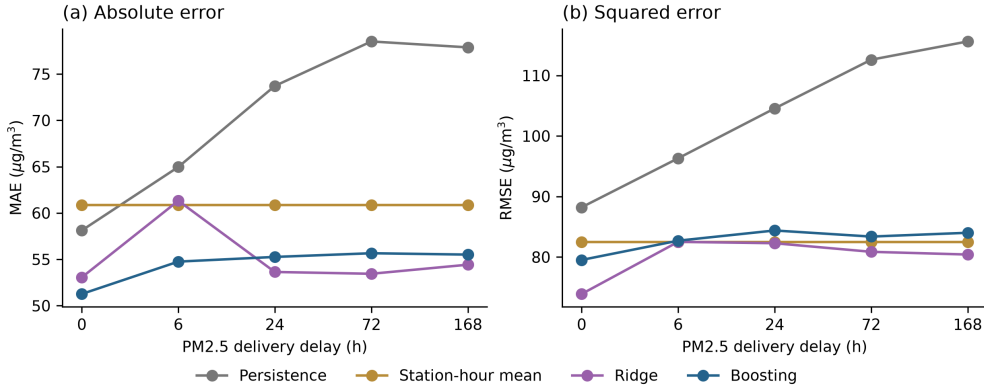


Fig. 1 Point-forecast errors under the five PM2.5 delivery delays. Models are trained once on development data and evaluated on identical test origins. The ordering of models depends on the error criterion and delay.

4.2 Interval performance under delay

At $d = 72$ hours and 90% nominal coverage, the static unmatched interval has coverage 88.15%, width 169.20 and score 377.14. Static matching changes these to 88.41%, 170.64 and 375.39. Rolling valid calibration attains coverage 88.32%, width 184.09 and score 328.01 (Table 4). Its average score is 13.03% lower than static unmatched calibration, but its intervals are wider and its coverage remains below 90%. Thus neither matching nor rolling updating establishes nominal coverage.

At 95% nominal coverage, static matching achieves 94.58% coverage and score 438.42, whereas rolling valid calibration attains 93.14% and score 391.54. Better average score therefore coexists with a larger coverage deficit. Across the five delays, rolling calibration improves the average score relative to the static unmatched reference, but the experiment does not demonstrate a universal ordering of calibration quality (Fig. 2).

Table 4 P: Original fixed-predictor comparison at a 72-hour delay. Coverage is expressed as a percentage; width and score are in $\mu\text{g}/\text{m}^3$. Each row uses 102,465 targets.

Nominal	Method	Coverage	Width	Score
90	Static unmatched	88.15	169.20	377.14
90	Static matched	88.41	170.64	375.39
90	Rolling valid	88.32	184.09	328.01
95	Static unmatched	93.59	221.15	458.87
95	Static matched	94.58	236.13	438.42
95	Rolling valid	93.14	225.21	391.54

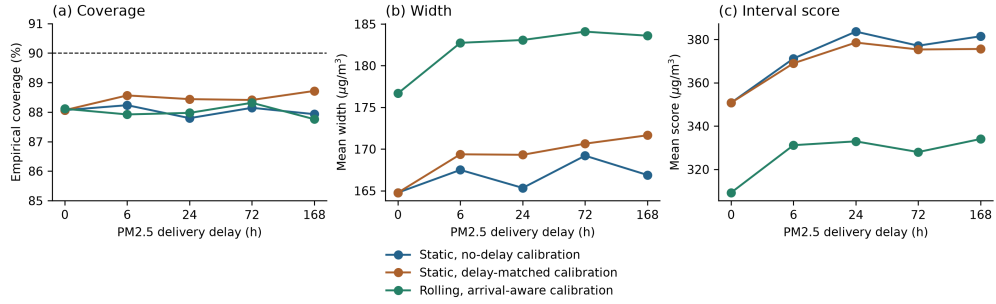


Fig. 2 Coverage, width and interval score at the 90% nominal level. Delay values are discrete experimental scenarios shown at equally spaced positions. The dashed line marks nominal coverage; all three approaches fall below it in aggregate.

4.3 Sensitivity within the original analysis (S)

The infeasible rolling diagnostic gives score 324.82 rather than 328.01 at 90% and a 72-hour delay. At 95% with a 168-hour delay it gives 385.01 rather than 399.95, making performance appear about 3.74% better if response arrival is ignored. These are descriptive differences from one test year. They do not justify assuming that premature feedback improves every metric or every subgroup.

For rolling valid intervals at 90% and $d = 72$, station-level coverage ranges from 85.60% at Dongsì to 91.93% at Huairou. Calendar-quarter coverage ranges from 81.91% in 2016Q4 to 95.29% in 2017Q1. The first and last calendar quarters are partial: 2016Q1 contains March only, and 2017Q1 contains January and February test origins. Subgroup results are descriptive and were not used to retune the evaluated models.

The primary paired score difference is -49.13. Its seven-day block-bootstrap interval excludes zero narrowly, but the intervals using 14- and 28-day blocks include zero (Table 5; Fig. 4). Consequently the result is suggestive of a lower average score in this sample, not robust evidence of superiority across reasonable block lengths. Treating 102,465 dependent test records as independent would not reflect this sensitivity.

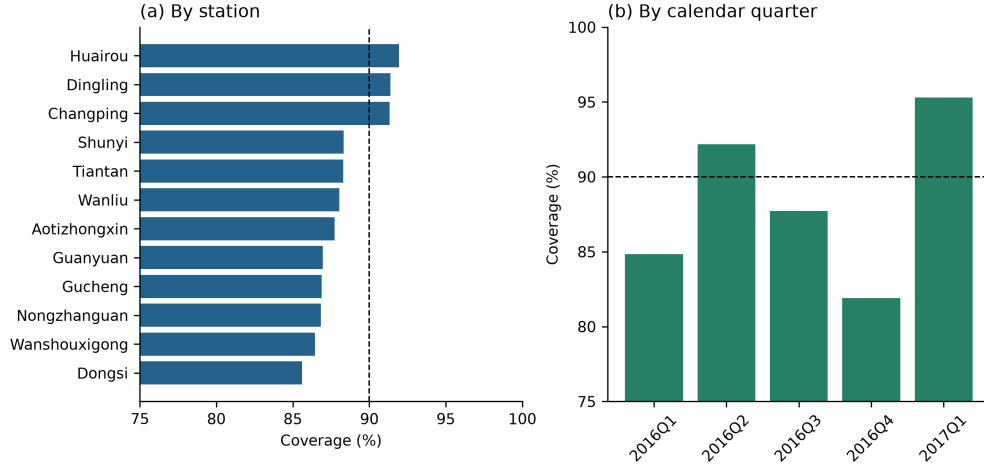


Fig. 3 Coverage heterogeneity for arrival-aware rolling intervals at a 72-hour delay and a 90% nominal level. The dashed lines indicate nominal coverage. Partial first and last quarters and unequal numbers of observed targets limit direct seasonal comparisons.

Table 5 S: Sensitivity of the primary paired score comparison to bootstrap block length. Negative values favour rolling valid calibration over static unmatched calibration.

Block (days)	Difference	95% percentile interval
7	-49.13	[-107.13, -3.02]
14	-49.13	[-118.54, 6.24]
28	-49.13	[-133.34, 3.60]

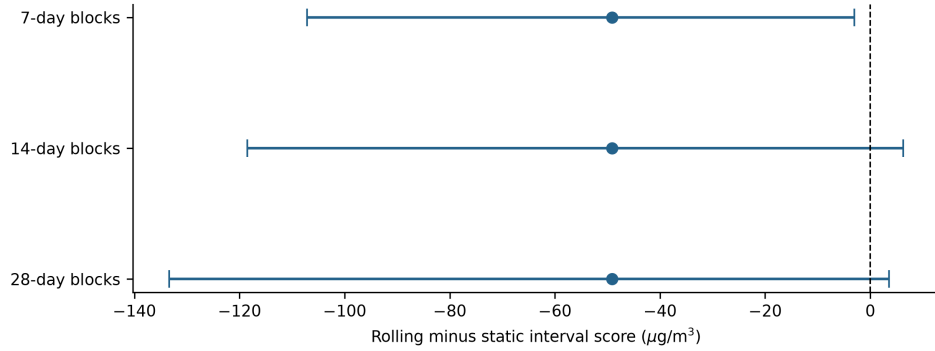


Fig. 4 Paired interval-score difference and block-bootstrap limits at 90% nominal coverage and a 72-hour delay. All estimates use the same observed targets and fixed fitted predictor.

5 Post-review exploratory extension

E: Exploratory comparisons

All added comparator analyses and their paired bootstrap intervals were specified after inspection of the principal test results. They cannot be promoted to independent confirmatory evidence.

5.1 Analysis status and training regimes

The following analyses were designed after the original test results had been inspected. Their separate local protocol and outputs preserve this chronology; they are exploratory additions, not a new preregistered test. No configuration is selected by final-test score. Experiment A retains the original fixed predictor. Experiment B selects 15 or 31 leaves, with 100 boosting iterations, using the same three expanding development folds but reconstructing predictors under the candidate delay. Training responses must have arrived before the relevant fold boundary. The final adapted fit uses only development responses arriving before 1 March 2015. Columns entirely unobserved in a training fold are excluded using that fold only. MAE summarizes absolute

Table 6 Information and selection regimes. The idealized reference is not deployable when measurements are delayed.

Regime	Training information	Selection and interpretation
A: fixed	Original, zero imposed delay	Original development choice; sensitivity to information loss
B: adapted	Same delay as evaluation	Delay-specific temporal validation; arrival-purged responses
Idealized	Zero imposed delay	Reference on the same targets; not a delayed deployment option

loss, whereas RMSE gives greater weight to large errors. The original boosting model is fitted with absolute-error loss; its MAE advantage does not establish superiority for decisions dominated by extremes. Station-level point metrics are supplied alongside Table 7.

5.2 Exact executed CQR algorithm

For each $d \in \{0, 6, 24, 72, 168\}$ hours and nominal error $\alpha \in \{0.10, 0.05\}$, let $X_t^{(d)}$ denote the causal delayed predictor vector. Four separate histogram-boosting quantile models $f_{d,p}$, $p \in \{.025, .05, .95, .975\}$, use pinball loss $\rho_p(u) = u\{p - \mathbb{I}(u < 0)\}$ (Koenker and Bassett 1978; Romano et al. 2019). Training uses development responses satisfying $t + 24 + d < 1$ March 2015, with training-only feature exclusion for entirely unobserved columns. Each quantile model has 15 leaves, 100 iterations, learning rate .05, minimum leaf size 40, L_2 regularization 1, no early stopping and seed 20261003.

Table 7 Fixed and delay-adapted point predictors. MAE and RMSE are in $\mu\text{g}/\text{m}^3$.

Delay	Fixed MAE	Adapted MAE	Fixed RMSE	Adapted RMSE
0	51.27	51.27	79.53	79.53
6	54.76	51.47	82.71	79.80
24	55.26	51.67	84.39	79.81
72	55.66	51.29	83.39	78.44
168	55.51	51.04	84.01	78.12

These settings are fixed: the 15/31-leaf model selection in Table 7 applies to the point predictor, not to these quantile models.

First clip each predicted quantile at zero, $g_{d,p}(x) = \max\{0, f_{d,p}(x)\}$, then order the pair:

$$\ell_d(x) = \min\{g_{d,\alpha/2}(x), g_{d,1-\alpha/2}(x)\}, \quad (6)$$

$$u_d(x) = \max\{g_{d,\alpha/2}(x), g_{d,1-\alpha/2}(x)\}. \quad (7)$$

The calibration set $\mathcal{C}$ comprises naturally observed calibration targets with measurement time strictly before 23 February 2016. The cutoff $c_\star = 23$ February 2016, 00:00 precedes the first test issue $t_\star = 1$ March 2016, 00:00 by exactly 168 hours (2016 is a leap year). Writing v_i for target measurement time, every retained response satisfies

$$v_i < c_\star, \quad 0 \leq d \leq 168 \text{ h} \implies r_i = v_i + d < c_\star + 168 \text{ h} = t_\star. \quad (8)$$

The latest retained hourly target is 22 February at 23:00, arriving at 29 February at 23:00 for $d = 168$ h, one hour before $t_\star$. Of the 103,000 naturally observed calibration responses surviving the original split rules, 1,988 are excluded by this common target-time cutoff. Therefore $n = |\mathcal{C}| = 101,012$ counts observed station-specific responses, with identical keys at every delay. No additional row is removed for missing predictors: the delayed reconstruction preserves rows, missing numeric values are handled by boosting, and categorical encodings are fixed from training. All 45 required source columns exist under every delay; entirely unobserved training columns can be excluded from the encoded design without deleting responses. The fitted designs retain 73, 70, 66, 64 and 62 encoded columns for delays 0, 6, 24, 72 and 168 h, respectively. These column exclusions do not change the response count. The supplementary sample/feature audit lists the retained encoded columns and a common hash of the calibration keys. There are twenty quantile learners (four per delay), combined into ten interval configurations (two nominal levels per delay); all use the same calibration-response exclusions. Compute the *signed* scores and rank correction:

$$s_i = \max\{\ell_d(X_i) - Y_i, Y_i - u_d(X_i)\}, \quad k = \lceil (n+1)(1-\alpha) \rceil, \quad Q_{d,\alpha} = s_{(k)}. \quad (9)$$

Here $s_{(k)}$ is the k th smallest score, without interpolation. Negative scores are retained; scores are neither absolute-valued nor clipped. The executed implementation asserts $1 \leq k \leq n$ and $Q_{d,\alpha} \geq 0$. The latter holds for every fitted configuration; inputs violating these assertions require an explicit methodological amendment rather than silent substitution. The issued CQR interval is

$$C_{t,d,\alpha} = \left[\max\{0, \ell_d(X_t^{(d)}) - Q_{d,\alpha}\}, u_d(X_t^{(d)}) + Q_{d,\alpha} \right]. \quad (10)$$

The nonnegative lower truncation uses the nonnegative concentration support. Both models and correction remain fixed throughout testing: test responses never recalibrate CQR. The raw quantile baseline uses $[\ell_d, u_d]$. At 72 hours, the corrections are 1.93848 and 1.25839 for 90% and 95%; their ranks are 90,912 and 95,963. All ten configurations and all stored test endpoints were reconstructed from the saved quantile models, with maximum discrepancy below 6×10^{-14} . Crossing counts were zero in this execution, although ordering remains part of the defined algorithm. The audit table supplies all corrections and arrival cutoffs. Serial dependence and distribution shift preclude importing an exchangeable-sample coverage guarantee. Contrasts with the fixed residual rules compare complete training-and-calibration pipelines, not calibration alone.

5.3 ACI-inspired heuristic: name and scope

We call the adaptive comparator the *ACI-inspired daily delayed-feedback heuristic (ADH)* (Gibbs and Candès 2021; Zaffran et al. 2022). It is not a reproduction of unmodified ACI and carries no transferred ACI guarantee. The historical machine identifier `ACI_delayed_projected` is retained only for compatibility with the published prediction files.

Let a_j be a daily midnight, r_i the response arrival time, and C_i^{issued} the interval actually issued for an earlier test forecast at t_i . Define

$$F_j = \{i : a_{j-1} < r_i \leq a_j, t_i < a_j, i \text{ is a test forecast}\}, \quad (11)$$

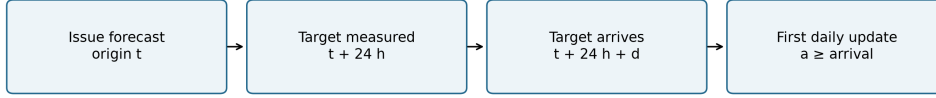
$$E_j = |F_j|^{-1} \sum_{i \in F_j} \mathbb{1}\{Y_i \notin C_i^{\text{issued}}\}, \quad (12)$$

$$\eta_j = \Pi_{[.001, .5]} \{\eta_{j-1} + .01(\alpha - E_j)\}. \quad (13)$$

Initialize $\eta_0 = \alpha$; skip the update when F_j is empty, including the first test update. Interval boundaries count as covered. A day receives one update regardless of its arriving batch size. Next take the rank-corrected $(1 - \eta_j)$ quantile of absolute residuals with arrivals in $(a_j - 30 \text{ days}, a_j]$, pooled across stations, and issue nonnegative-truncated symmetric intervals around the fixed point predictor for that day. Arrivals at the exact midnight are processed before issuance. Daily batching, unequal batch sizes, projection, a rolling residual window and delayed feedback differ from the original ACI setting. Neither gain .01 nor bounds $[.001, .5]$ were optimized on the test. These choices define an exploratory heuristic; they establish neither long-run nor conditional coverage. SPCI, EnbPI and missing-data augmentation remain unimplemented.

5.4 Information-set audit

For constant delay d , a forecast issued at t concerns Y_{t+24} , whose arrival is $t + 24 + d$. The first permitted daily update is the next midnight at or after arrival, assuming arrivals at that exact instant are processed before issuance. At each update, the residual set contains only arrivals in $(a - 30 \text{ days}, a]$. The fitted predictors remain fixed; only the calibration quantile and, for ADH, the error-level state evolve. Event ledgers



At update a : use only residuals with arrival $\leq a$; keep fitted predictors fixed.

Fig. 5 Measurement, arrival and update are distinct events. The same delay restricts predictors and feedback.

retain issue time, target time, arrival, first inclusion, last possible use and whether the residual is used within the evaluated period. Daily audit tables record the latest eligible arrival. Independent checks reconstruct selected quantiles and perturb unavailable residuals to verify invariance. For heterogeneous delays, eligibility uses each station's own arrival time rather than a shared measurement cutoff.

Table 8 E: Exploratory pipeline trade-offs at 72 hours. CQR denotes CQR with delay-specific training. Coverage and its pointwise 95% circular 14-day block-bootstrap interval are percentages (1,000 replicates; no simultaneous multiplicity adjustment). Width and score are in $\mu\text{g}/\text{m}^3$.

Nominal	Method	Coverage	Coverage interval	Width	Score
90	CQR	88.91	[86.51, 91.18]	182.72	258.40
90	ADH	89.78	[87.15, 92.42]	203.17	317.48
90	Rolling, by station	88.32	[85.30, 91.12]	184.94	324.86
90	Rolling, equal weights	88.32	[85.23, 91.16]	184.08	328.05
95	CQR	93.82	[92.03, 95.44]	211.01	307.18
95	ADH	94.64	[92.68, 96.37]	253.79	367.12
95	Rolling, by station	93.28	[91.10, 95.30]	225.69	386.58
95	Rolling, equal weights	93.13	[90.85, 95.21]	225.16	391.70

At 90%, CQR's observed coverage is below nominal. Its dependence-aware interval [86.51,91.18]% contains 90%, so the point estimate alone does not establish a distinguishable departure from nominal; neither does interval inclusion prove calibration.

CQR has the lowest observed score among the evaluated pipelines at 72 hours, at both nominal levels. ADH is closer to nominal coverage but produces wider intervals and a larger score. These are distinct criteria: coverage measures inclusion, width describes concentration, and the chosen score penalizes both width and misses. No uniform or application-independent superiority follows.

5.5 Paired score uncertainty for exploratory contrasts

The exploratory paired contrasts use 10,000 replicates to reduce Monte Carlo error in percentile limits; the original protocol retains its prespecified 1,000-replicate analysis. The previously reported exploratory coverage and subgroup intervals also retain 1,000 replicates and seed 20261005 + L . A larger replicate count improves numerical precision of the bootstrap approximation, not its assumptions or evidential status. For the direct comparisons, 10,000 circular day-block draws resample paired score sums and observed-target counts together, retaining all stations per day. Blocks are 7, 14 or 28 days; the random seed is 20261006 + L for block length L . Each bootstrap mean is the resampled total difference divided by the resampled total target count, preserving the observation-weighted estimand. Pointwise percentile limits are the .025 and .975 quantiles. These intervals condition on stored fitted models; they include neither model-selection/refitting uncertainty nor simultaneous multiplicity control. Circular joining and nonstationarity remain limitations. This separate extension does not replace the original primary bootstrap.

Table 9 E: Exploratory paired score differences at 72 hours. Pointwise 95% block-bootstrap intervals; no simultaneous multiplicity adjustment. Negative values favour the first method. Units: $\mu\text{g}/\text{m}^3$.

Nominal	Contrast	Difference	7-day interval	14-day interval	28-day interval
90	CQR – rolling	-69.61	[-112.27, -32.38]	[-113.16, -30.66]	[-119.36, -24.91]
90	ADH – rolling	-10.53	[-31.74, 4.70]	[-30.98, 5.24]	[-33.94, 6.71]
90	CQR – ADH	-59.07	[-99.96, -21.60]	[-99.84, -21.36]	[-101.59, -22.69]
95	CQR – rolling	-84.36	[-157.06, -21.09]	[-156.06, -21.72]	[-155.84, -22.91]
95	ADH – rolling	-24.42	[-59.11, 3.36]	[-56.75, 2.90]	[-63.42, 5.41]
95	CQR – ADH	-59.95	[-125.99, 6.27]	[-125.36, 1.90]	[-120.84, -10.60]

This table compares entire pipelines: CQR changes the fitted quantile learner, delay-specific training, interval shape and calibration relative to the original fixed-point-predictor rolling rule. It does not isolate the causal effect of conformal calibration or predictor adaptation. At 90%, the descriptive score reduction is $69.61/328.01 \approx 21.2\%$. Here rolling means the original global rolling-valid rule (score 328.01), not the equal-station-weight sensitivity (328.05).

At 90%, CQR minus rolling has mean difference -69.61 and 14-day interval $[-113.16, -30.66]$. ADH minus rolling has difference -10.53 with interval $[-30.98, 5.24]$; an adaptive-heuristic advantage is therefore not established by this

comparison. CQR minus ADH is -59.07 , with interval $[-99.84, -21.36]$. Even intervals excluding zero remain post-hoc, exploratory evidence from the same inspected test period. The supplementary contrast table includes all delays, both nominal levels, static baselines, raw quantile intervals, and separately labelled earlier-period and heterogeneous-delay sensitivities.

6 Sensitivity analyses and scope diagnostics

S: Sensitivity evidence

These checks characterize dependence assumptions, pooling, missingness and scope. The original block-length sensitivity is preserved; post-review sensitivities use already inspected data. The earlier Beijing period is not independent external validation.

6.1 Coverage uncertainty and station pooling

The supplementary tables report coverage, paired coverage differences and paired score differences using 1,000 circular day-block bootstrap replicates at block lengths 7, 14 and 28. Each resampled day retains all stations. For station and quarter comparisons at 72 hours, exploratory simultaneous bands use the bootstrap maximum absolute deviation of coverage differences within each method-level subgroup family. These do not establish global multiplicity control across every reported analysis. Replicates without observations in every subgroup are excluded and their count is reported. Conditional-on-fit inference and nonstationarity remain limitations.

For the original rolling rule at 90%, the 14-day coverage interval is $[85.23, 91.17]\%$; its coverage difference from static unmatched is 0.17 percentage points, with interval $[-3.54, 3.94]$. Figure 6 shows that coverage estimates should not be interpreted without dependence-aware uncertainty. Inclusion of nominal coverage in an approximate interval is not proof of calibration.

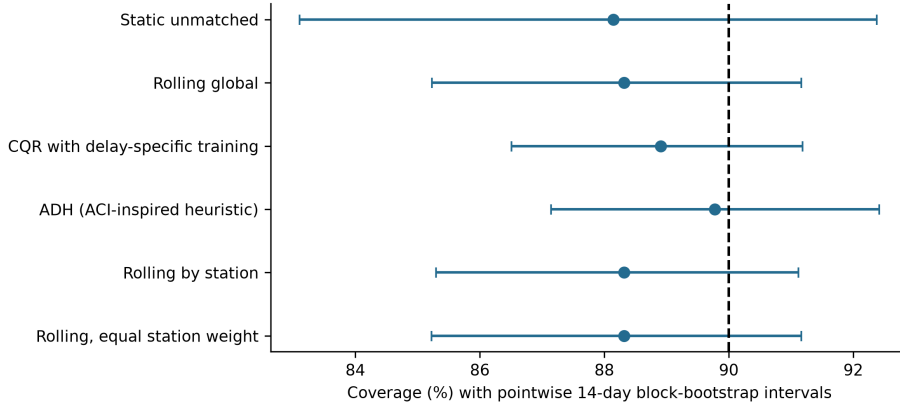


Fig. 6 Coverage uncertainty at 72 hours and 90% nominal level. Intervals are pointwise 95% circular 14-day block-bootstrap intervals, conditional on fitted models.

Pooling is examined using a common global residual quantile, separate station quantiles, and a weighted empirical global quantile giving each station equal total weight within the active window. The weighted rule is a sensitivity analysis, not a claim of exact finite-sample conformal validity. More observations give a station greater influence under ordinary pooling, whereas equal total weights remove that particular imbalance. All three methods retain temporal and cross-station dependence.

6.2 Practical loss, missingness and scope

In the original 72-hour, 90% comparison, rolling improves the daily mean score on 56.6% of days and produces wider intervals for 47.3% of targets. It newly covers 4,217 targets but loses coverage of 4,043, a net change of 174. The mean width change is $14.88 \mu\text{g}/\text{m}^3$, and the change in the score's exclusion penalty is -64.01. Thus the score improvement cannot be equated with a uniform gain in inclusion. Daily score-difference quantiles, concentration-stratified comparisons and hypothetical width-plus-exclusion-distance losses are provided. These losses are not monetary costs or validated operational utilities.

Missingness is quantified by station, calendar month, hour, study period and development-defined quintiles of PM10, NO2, temperature, pressure and wind speed. All absent PM2.5 records are retained in this audit. Concentration-stratified performance uses development-defined response quantiles and is retrospective; the target concentration is unknown at issuance. The audit does not identify the missingness mechanism. Observation-probability weighting is not reported as a correction because it would require an additional observation model, positivity assessment and unverifiable assumptions about absent outcomes. Population-wide performance remains unidentified without such assumptions.

One additional synthetic scenario assigns delays 6, 24, 72 and 168 hours cyclically to alphabetically ordered stations. This tests heterogeneous availability across the network; it does not reproduce intermittent, out-of-order or concentration-dependent delivery. An earlier temporal backtest trains on responses arriving before January 2014, calibrates on later development records before 22 February 2015, and evaluates 103,000 targets in March 2015–February 2016 at a 72-hour delay. At 90%, static matched, rolling, ADH and CQR have scores 306.23, 296.79, 295.23 and 263.41, respectively; their coverages are 92.69%, 89.66%, 89.79% and 92.94%. The model configuration was fixed for this sensitivity. Its data overlap the broader study's development/calibration history, so this is neither external replication nor independent confirmation.

7 Discussion

Three observations stand out. First, temporal validation does not support a simple claim that more complex boosting configurations are preferable. Ridge and boosting trade places depending on the metric and delay. This argues for retaining transparent baselines and several error criteria when reporting an environmental forecasting model.

Second, the central difficulty is not only the loss of a recent predictor. The available residual sample also changes when the target channel arrives late. The diagnostic

results quantify an apparent improvement that can arise when this constraint is ignored, especially at long delays and higher nominal coverage. A reproducible benchmark should therefore specify an arrival time for every measurement entering its update rule, not merely avoid future covariates in the original feature matrix.

Third, calibration choices involve trade-offs. Rolling residuals produce better average scores here, but their 95% coverage is lower than the delay-matched static method. Static matching partly compensates for predictor degradation without guaranteeing nominal coverage. Strong variation among quarters and stations further cautions against treating the network-wide average as a property shared by every location or future period. These findings are consistent with the broader distinction between marginal and more local notions of coverage in missing-data inference (Zaffran et al. 2023).

The prespecified objective tests whether recalibrating a fixed predictor with valid delayed feedback changes score relative to static unmatched calibration. The exploratory objective asks how alternative complete pipelines change that ranking. At 72 hours, CQR with delay-specific training has the lowest observed score among evaluated pipelines; ADH is closer to nominal coverage at the cost of greater width. These findings do not identify a universally preferable method. A lower score for rolling relative to a static residual rule does not establish it as the best evaluated pipeline. Conditional quantile estimation and training adaptation must be considered alongside calibration. The experiment still does not compare the full range of time-series or missing-data methods (Xu and Xie 2023; Zaffran et al. 2023).

Several design choices constrain interpretation. Simulated constant delays do not reproduce intermittent network failures, outcome-dependent missingness, or co-occurring loss of other channels. Reusing current correlated pollutants can soften the effect of delayed PM2.5. Natural missing responses are not recovered, so scoring is conditional on observation. There is only one city, one principal final test year and an overlapping earlier temporal sensitivity. Delay-adapted training improves the design but does not exhaust the attainable model space. All post-review comparisons are exploratory because the principal test was already examined.

The bootstrap intervals condition on the fitted predictor and assume that resampling time blocks gives a useful approximation despite seasonal changes. They omit uncertainty from model selection and refitting, and cannot turn twelve correlated stations into independent regional replications. Sensitivity to block length is therefore a substantive result rather than a technical detail. The evidence is insufficient to claim general improvement, operational readiness, or a new distribution-free guarantee.

8 Conclusion

The locally prespecified comparison shows a lower average interval score for rolling-valid than static-unmatched calibration under the original fixed predictor, with 88.32% coverage at 90% nominal and a superiority conclusion sensitive to bootstrap block length. This result addresses feedback-aware recalibration, not the optimal forecasting pipeline.

The post-review exploration shows that delay-specific training and conditional quantile estimation can substantially change the observed ranking. At 72 hours, CQR with delay-specific training has the lowest observed score among the evaluated pipelines, while ADH is closer to nominal coverage with wider intervals. CQR still undercovers in its point estimate, and approximate intervals do not establish calibration. These are whole-pipeline, exploratory contrasts on an already inspected test set. The contribution is an auditable benchmark that separates information availability, score, coverage and width. External replication, independent confirmation and application-specific loss assessment remain necessary before deployment recommendations.

Supplementary information

The local research package contains the data-preparation script and original archive, chronological split indices, the locally recorded protocol, development scores, all test predictions, subgroup and bootstrap tables, and the independent R verification script. The research materials are provided through the GitHub repository listed below; the frozen analysis-code DOI is listed below.

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Statements and Declarations

Data and code availability

The original UCI source data are available at <https://doi.org/10.24432/C5RK5G> under CC BY 4.0. The fixed analysis-code snapshot v0.3.0, including environment requirements, local protocols, method documentation, checksum manifest and execution instructions, is deposited as software at <https://doi.org/10.5281/zenodo.23171630>. This Zenodo record contains code and documentation only: it does not contain the manuscript, original or derived data, fitted models or prediction files.

Those materials are available in the public GitHub research releases at <https://github.com/rntvargas/beijing-pm25-delayed-feedback/releases>. The complete `paper_springer_revision_reproducible.zip` contains the source archive and derived partitions in `outputs/02_datos_estudio`; fitted models, predictions, analysis configurations and result tables in `outputs/03_experimento` and `outputs/05_revision`; and the manuscript, figures, bibliography and Springer source in `outputs/04_manuscrito_springer`. The smaller `source_and_tables.zip` contains source, documentation and tables. The executed code and numerical results correspond to research release v0.3.0 (commit `a8364ec884bbf45bf754053762d935497a68125f`); editorial release v0.3.1 adds this clarification and a sample/provenance audit without changing fitted models or predictions. Archive manifests identify exact file checksums. The software DOI is not an article DOI or an archival DOI for the complete data/model package. The UCI and third-party template licenses remain distinct from the MIT license for original analysis code.

Use of artificial intelligence

OpenAI Codex desktop assisted during 2–6 October 2026 with literature search and organization, study-design development, programming, computational execution, automated verification, figure preparation from numerical outputs, and manuscript drafting/editing. Available session records identify model `gpt-6-astra`; an immutable backend build identifier was not recorded. Numerical analyses were run through Python/R tools and are supported by archived code and outputs; figures were plotted

from those outputs rather than generated as unverifiable illustrations. The automated reconstruction and R checks reported here are computational checks, not evidence of independent personal verification by the author. The AI system is not an author. Before submission, the author must review and approve the analyses, references, figures and claims, confirm the extent of personal verification, and accept responsibility for the final content. Those author confirmations remain outstanding in this research draft; they are not inferred from the use of the tools.

Authorship, funding and competing interests

Renato Quispe Vargas, Universidad Nacional del Altiplano, supplied the authorship and affiliation information. Correspondence details, CRediT contributions, funding and competing-interest declarations remain to be confirmed. No absence of funding or conflicts is presumed.

Ethics

The analysis uses publicly distributed environmental measurements and contains no participant-level health records. Any institution-specific determination or statement required by the selected journal remains for the authors to confirm.